Spring 2026 Section: _____ Name:_____

Read the following directions carefully. DO NOT turn to the next page until the exam has started.

Write your name and section number at the top right of this page:

Class	Section Number
Miranda 8:50	001
Miranda 9:55	002
Sahifa 1:20	003
Sahifa 2:25	004

As you complete the exam, write your initials at the top right of each other page.

When the exam start time is called, you may turn the page and begin your exam.
If you need more room, there is a blank page at the end of the exam, or we can give you some scratch paper.

Some multiple choice questions are “Select ONE" while others are “Select ALL that apply". Pay attention to the question type and only mark one option if it says “Select ONE". Fill in the circles completely.

If you finish early, you can hand your exam to your instructor or TA and leave early.

Otherwise, stop writing and hand your exam to your instructor or TA when the exam stop time is called.

Some of the following problems are based on the `baseball` dataset, which records batting, pitching, and fielding statistics for 50 players in a community baseball league. Each player, identified by `id`, has batting statistics recorded. The `position` column identifies whether the player has fielding (FIE) or pitching (PIT) statistics recorded. The first six rows and a brief description of the statistics are given below.

```
id   BAT_RBI   BAT_AVG   position   FIE_DBP   FIE_ERR   PIT_AVG   PIT_SHO
 2         4     0.126    infield        53         5        NA        NA
35         4     0.124    pitcher        NA        NA     7.026         1
50         7     0.174    pitcher        NA        NA     7.166         0
 3         8     0.250    infield        66        15        NA        NA
33         4     0.142    infield        91         9        NA        NA
19         4     0.128   outfield        59        14        NA        NA
.... plus 44 more rows
```

- `BAT_RBI`: runs batted in
- `BAT_AVG`: overall batting average
- `FIE_DBP`: double-plays earned
- `FIE_ERR`: fielding errors made
- `PIT_AVG`: average strikeouts per game
- `PIT_SHO`: shutout games (no runs scored)

1. Your friend is trying to work with the `baseball` data but needs help debugging. Each part shows what your friend was trying to do and the code they used, which either returns an error or an undesirable result. For each part, identify the error / incorrect result that will output and suggest a fix. There may be more than one possible fix. Assume `baseball` is the only object that exists in the R environment.

   (a) (2 points) We want to calculate the average of the `PIT_AVG` column for all pitchers in the data.

   ```
   mean(baseball$PIT_AVG)
   ```

   - What is the output?
     Solution: The output will be `NA`.
   - How can we fix it?
     Solution: Either filter out the `NA` values or set `na.rm = T` inside the `mean()` function.

   (b) (2 points) We want to count the number of infield players is the data.

   ```
   sum(baseball$position == infield)
   ```

   - What is the output?
     Solution: The output will be an error with the object `infield` not found.
   - How can we fix it?
     Solution: To identify the category correctly, we want to have `"infield"` in quotes to make it a string.

   (c) (2 points) We want to find the player with the highest batter average.

   ```
   max(baseball$3)
   ```

   - What is the output?
     Solution: There will be an error about the `‘3‘` column not being found.
   - How can we fix it?
     Solution: We can use `baseball[3]` to get the third column of the data or identify it by name with `baseball$BAT_AVG`.

2. Continue working with the `baseball` data from problem 1.

```
id   BAT_RBI   BAT_AVG   position   FIE_DBP   FIE_ERR   PIT_AVG   PIT_SHO
 2         4     0.126    infield        53         5        NA        NA
35         4     0.124    pitcher        NA        NA     7.026         1
50         7     0.174    pitcher        NA        NA     7.166         0
 3         8     0.250    infield        66        15        NA        NA
33         4     0.142    infield        91         9        NA        NA
19         4     0.128   outfield        59        14        NA        NA
.... plus 44 more rows
```

The following players are eligible to win an award from the baseball league:

- (A) Any player with a batting average above 0.28
- (B) Any pitcher with at least one shutout game (`PIT_SHO`)
- (C) Any non-pitcher (infield or outfield) with more than 70 double plays and fewer than 10 errors

Consider using R to identify these players in the data. For example, to return rows who meet condition (A) we can use:

```
baseball %>% filter(BAT_AVG > 0.28)
```

(a) (4 points) What inputs can go into `filter()` to find rows that meet condition (B)? **Select ALL that apply**.

- ● `position == "pitcher" & PIT_SHO > 0`
- ● `position == "pitcher" & PIT_SHO != 0`
- ○ `!is.na(PIT_SHO)`
- ○ `position == "pitcher" | PIT_SHO > 0`

(b) (4 points) What inputs can go into `filter()` to find rows that meet condition (C)? **Select ALL that apply**.

- ○ `position %in% c("infield", "outfield") & (FIE_DBP > 70 | FIE_ERR < 10)`
- ● `position %in% c("infield", "outfield") & (FIE_DBP > 70 & FIE_ERR < 10)`
- ● `position != c("pitcher") & (FIE_DBP > 70 & FIE_ERR < 10)`
- ○ `position != c("pitcher") | (FIE_DBP > 70 & FIE_ERR < 10)`

(c) (4 points) The dataframe `past_awardees` contains the 11 players who have previously won an award, identified by the same `id` column as in `baseball`. How can the `baseball` data be joined with the `past_awardees` to ensure that the rows in the output are all players who have **not** received an award before?

You should identify **which** join function to use as well as the **order** of the two datasets, and briefly explain your answer.

Solution: We should use `anti_join(baseball, past_awardees)`. This will get us a filtered version of `baseball` where the output is all rows that do not appear in `past_awardees` (i.e. players who have not already won an award).

3. Consider the following dataframe that shows the name, height (in inches), and weight (in pounds) of three 5-year-old children.

name	measurement	value
Ross	height	43
Ross	weight	40.2
Terry	height	46
Terry	weight	43.6
Ellie	height	44
Ellie	weight	39

(a) (5 points) We want to make a new column called `metric_value` that gives height in centimeters (inches times 2.54) and weight in kilograms (pounds times 0.45). Fill in the blanks of `case_when` below to create this column.

```
data %>%
    mutate(___metric_value___ = case_when(
        measurement == "height" ~ ___value * 2.54___,
        measurement == "weight" ~ ___value * 0.45___))
```

(b) (4 points) Return to the original dataframe. Which statements are true about the output of the `pivot_wider()` code below? **Select ALL that apply**.

```
wide_data <- pivot_wider(data, names_from = measurement, values_from = value)
```

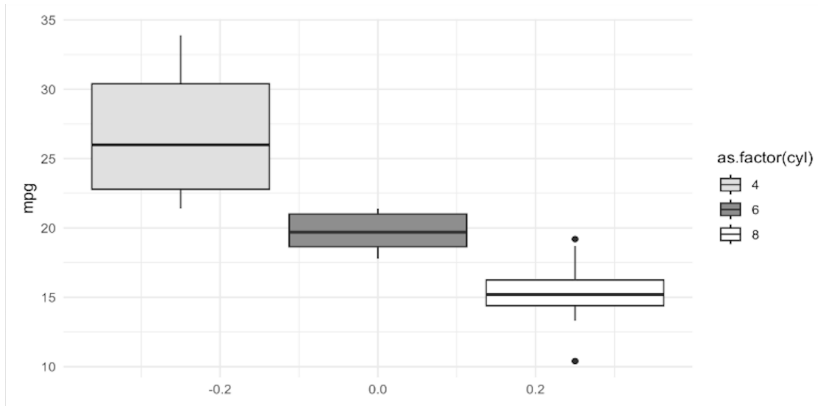
- ● The output `wide_data` has **fewer** rows than the original data.
- ○ The output `wide_data` has **more** columns than the original data.
- ○ Some of the information in the original data was lost and had to be replaced with `NA`.
- ● There are multiple numeric-type columns in the output `wide_data`.

(c) (4 points) Consider doing the same metric conversion in (a) but using the wide-form data made in (b). Fill in the blanks to calculate the average height and weight of the three children.

```
wide_data %>% ___summarize___(
    avg_height = ____mean____(___height___),
    avg_weight = ____mean____(___weight___))
```

4. Below is the code and plot that depicts the distribution of the variable `mpg` of `mtcars` dataset.

```
ggplot(mtcars, aes(y = mpg)) +
  geom_boxplot(aes(fill = as.factor(cyl)))
```



(a) (4 points) Using the code above, identify which inputs in the code corresponds to each type of aesthetic listed below. Some inputs can be used for multiple blanks, and some blanks may be left empty.

Variable aesthetic: `y = mpg` and `fill = as.factor(cyl)`

Constant aesthetic: _____

Global aesthetic: `y = mpg`

Local aesthetic: `fill = as.factor(cyl)`

(b) (3 points) Which of the following statements correctly describes the box-plot? Select **ALL** that apply.

- ○ There are no outliers in `mpg` for cars with 8 cylinders.
- ● The distribution of `mpg` for cars with 4 cylinders is skewed (non-symmetric).
- ● `mpg` for cars with 8 cylinders has one outlier on both sides.
- ● The 25th percentile of `mpg` for cars with 4 cylinders is approximately equal to 23.
- ● `mpg` for cars with 6 cylinders has no outliers.
- ● The median of `mpg` for cars with 8 cylinders is approximately equal to 15.

The Language Fluency dataset, saved in the environment as `Language`, contains student names, each parent's first language, and indicator variables for five languages. Each language column specifies whether the individual speaks that language (e.g., TRUE/FALSE).

```
A tibble: 50 × 7
Name           Mom_speaks Dad_speaks English Chinese French Arabic Spanish
<chr>          <chr>      <chr>      <lgl>   <lgl>   <lgl>  <lgl>  <lgl>
Henry Todd     English    English    TRUE    FALSE   FALSE  FALSE  TRUE
Ren Xiao       Japanese   Chinese    TRUE    TRUE    FALSE  FALSE  FALSE
Wendy Li       Chinese    English    TRUE    FALSE   FALSE  FALSE  TRUE
Taylor Leight  English    English    TRUE    FALSE   TRUE   FALSE  TRUE
Joseph Lopez   Spanish    English    TRUE    FALSE   FALSE  FALSE  FALSE
Aaron Sato     English    Japanese   TRUE    TRUE    FALSE  FALSE  FALSE
.... plus 44 more rows
```

The *Undergraduate Major and GPA* dataset saved in the environment as `Major` contains the names of 20 students (15 of them are also present in the `Language` dataset), their Undergraduate majors, and other columns.

```
A tibble: 20 × 5
Student_Name        Undergrad_major         Year_of_graduation   GPA Grad_School
<chr>               <chr>                             <dbl> <dbl> <chr>
Henry Todd          Computer Science                   2020  3.45 YES
Monica Keale        Biology                            2021  2.78 NO
Wendy Li            Psychology                         2027  3.1  NA
Taylor Leight       Business Administration            2022  3.92 YES
Ezequial McPherson  English Literature                 2024  3.88 NO
Joseph Lopez        Mathematics                        2026  3.98 YES
.... plus 14 more rows
```

5. Use the `Language` and `Major` to answer the following questions.

(a) (3 points) A hiring agency is interviewing the students and wants to only interview students whose language fluency data is available. Use a mutating joins and fill in the blanks to join the datasets such that only the 15 common students are present in the final dataset.

```
____inner_join____(Major, Language,
        ________by________ = join_by(Name == Student_Name))
```

(b) (2 points) The agency also wishes to convert the `Language` dataset into something more easily readable by putting the language information next to their names. The dataset should only contain the following information: Name, English through Spanish. Select the code that will then put all the languages beside the names in a column called `Language` and a column to denote whether the person knows that language or not called `Fluent`. **Select ONE**.

○ `Language %>% relocate(Name, English:Spanish) %>%`
`pivot_longer(c(English, Spanish), names_to = "Language", values_to = "Fluent")`

● `Language %>% select(Name, English:Spanish) %>%`
`pivot_longer(English:Spanish, names_to = "Language", values_to = "Fluent")`

○ `Language %>% select(Name, English:Spanish) %>%`
`pivot_wider(names_from = "Language", values_from = "Fluent")`

○ `Language %>% relocate(Name, English:Spanish) %>%`
`pivot_wider(c(English, Spanish))`

6. (4 points) Match each scenario described below with the most appropriate ‘ggplot2‘ function. Choose from the list of functions provided. You may not need to use every one.

`[geom_smooth, geom_bar, geom_line, geom_col, geom_density, geom_histogram, geom_point]`

a A student is analyzing the noble gases and the number of known isotopes for each element, as shown in the table below. She wishes to display the number of isotopes of all 7 noble gases as bars.

Noble Gas	No. of isotopes
Helium	2
Neon	3
Argon	3
Krypton	6
Xenon	9
Radon	0
Oganesson	1

She should use the function _____`geom_col`_____.

b A scientist wishes to understand the distribution of BMI of the 300 participants in an experiment by plotting a curve for the density. He should use the function _____`geom_density`_____.

c An airport authority is trying to figure out the most affected flight carriers during COVID in 2020 by looking at a data set from Chicago airport that lists all flights and their carriers (airline name) departing from the airport. He should use the function _____`geom_bar`_____ since it can automatically count the number of flights for each distinct carrier in the data set.

d A school science teacher wishes to understand the distribution of the scores of his class by making a plot that shows the number of students in each score interval ([90 - 100], [80-90], and so on). The function _____`geom_histogram`_____ should be used since the x-axis in this plot is divided into bins or intervals.

7. (4 points) Identify whether the statements given below are TRUE/FALSE.

(a) The `slice_max()/slice_min()` functions can only be used inside `summarize()`.

○ TRUE ● FALSE

(b) `geom_density()` function can be used to make a density plot of a categorical variable.

○ TRUE ● FALSE

(c) When using the `seq()` function, we always need to specify the arguments "`from`", "`to`", and "`by`."

○ TRUE ● FALSE

(d) The default of the `method` argument of `geom_smooth()` function is `"lm"`.

○ TRUE ● FALSE

(e) The first argument of any join or pivot function is the dataset by default, unless specified otherwise.

● TRUE ○ FALSE

(f) A variable can be saved as '`2x`' in the environment.

○ TRUE ● FALSE

(g) `right_join()` returns all rows of the right dataset and all columns from both datasets.

● TRUE ○ FALSE

(h) The pipe operator can be used with other packages like `ggplot` to pipe things into the functions as the first argument.

● TRUE ○ FALSE